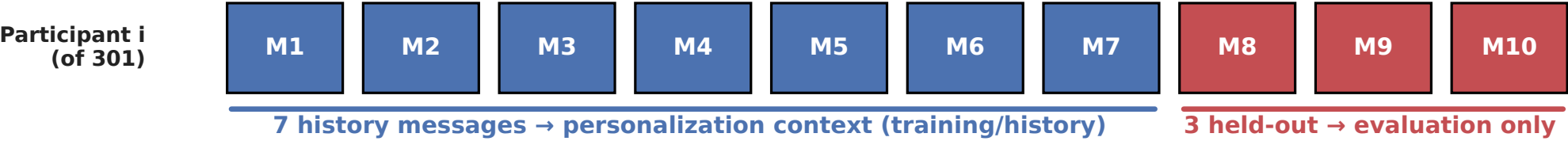


Within-participant 70/30 partition · identical for every method



What each method sees: same partition, different inputs → apples-to-apples comparison

Supervised baseline (Random Forest)	LLM-PP (persona + history prompt)	Hybrid RF + PP (RF prior + LLM update)
INPUTS - Participant demographics (age, sex, race, ...) - Avg of participant's 7 prior ratings — no per-message text or context —	INPUTS - Participant demographics 7 history (message, rating) pairs - Held-out message text	INPUTS - Everything above, plus - RF predicted class injected into the digital-twin prompt as a soft prior
OUTPUT Predicts each held-out message's rating from tabular features only	OUTPUT Predicts held-out rating via in-context inference (no model fine-tuning)	OUTPUT Predicts held-out rating with RF anchor + LLM refinement step
[✓] Same 3 held-out messages per participant <i>Trained across all 301 participants' 7-history items</i>	[✓] Same 3 held-out messages per participant <i>Each held-out call is independent, per participant</i>	[✓] Same 3 held-out messages per participant <i>Combines tabular + textual personalization</i>

Leakage audit · source: revision/figures/label_leakage_analysis.csv

Few-shot exemplar overlap with test	0 / 2 (0.0%)	no overlap
Within-participant duplicate items (history ≈ test)	16 / 319 (5.0%)	removed before evaluation (n=302-303)
Held-out items inserted into prompt history	0 / 0 (0.0%)	programmatic ID-based assembly
Cross-split message text reuse	104 / 107 (97.2%)	by design: same message text, different participants